

Akifumi Wachi

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RESEARCH INTERESTS

My research interests lie primarily in reinforcement learning, and span the entire theory-to-application spectrum from foundational advances all the way to deployment in real systems. I am particularly enthusiastic about conducting both empirical and theoretical studies that contribute to the improvement of reinforcement learning, ensuring its efficacy, safety, and reliability in practical applications.

WORK EXPERIENCE

Senior Chief Research Scientist LY Corporation	May 2025 - present
Chief Research Scientist LY Corporation	Oct 2023 - Apr 2025
Senior Research Scientist LINE Corporation	Sep 2022 - Sep 2023
Research Scientist IBM Research AI	Apr 2018 - Aug 2022

EDUCATION

Mar 2021 Ph.D. in Computer Science at University of Tsukuba
Mar 2018 M.S. in Aeronautics and Astronautics at University of Tokyo
Mar 2016 B.S. in Aeronautics and Astronautics at University of Tokyo

PUBLICATIONS

1. Shokichi Takakura, Akifumi Wachi, Rei Higuchi, Kohei Miyaguchi, Taiji Suzuki “Inference-Aware Meta-Alignment of LLMs via Non-Linear GRPO” International Conference on Machine Learning (ICML), 2026.
2. Mikoto Kudo, Takumi Tanabe, Akifumi Wachi, Youhei Akimoto “Sample-Efficient Hypergradient Estimation for Decentralized Bi-Level Reinforcement Learning” International Conference on Automated Planning and Scheduling (ICAPS), 2026.
3. Shigeki Kusaka, Keita Saito, Mikoto Kudo, Takumi Tanabe, Akifumi Wachi, Youhei Akimoto, “Cost-Minimized Label-Flipping Poisoning Attack to LLM Alignment” AAAI Conference on Artificial Intelligence (AAAI), 2025.
4. Akifumi Wachi, Kohei Miyaguchi, Takumi Tanabe, Rei Sato, Youhei Akimoto, “A Provable Approach for End-to-End Safe Reinforcement Learning” Neural Information Processing Systems (NeurIPS), 2025.
5. Runze Yan, Xun Shen, Akifumi Wachi, Kohei Miyaguchi, Takumi Tanabe, Rei Sato, Youhei Akimoto, Sebastien Gros, Anni Zhao, Xiao Hu, “Offline Guarded Safe Reinforcement Learning for Medical Treatment Optimization Strategies” Neural Information Processing Systems (NeurIPS), 2025. (Spotlight)
6. Kensuke Tatematsu, Akifumi Wachi, “Target Return Optimizer for Multi-Game Decision Trans-

former” Asian Conference on Machine Learning (ACML), 2025.

7. Kazumune Hashimoto, Yuga Onoue, Akifumi Wachi, Xun Shen, “Learning-Based Event-Triggered MPC With Gaussian Processes Under Terminal Constraints” IEEE Transactions on Cybernetics, 2025.
8. Akifumi Wachi, Thien Q. Tran, Rei Sato, Takumi Tanabe, Youhei Akimoto, “Stepwise Alignment for Constrained Language Model Policy Optimizations” Neural Information Processing Systems (NeurIPS), 2024.
9. Xun Shen, Shuo Jiang, Akifumi Wachi, Kazumune Hashimoto, Sebastien Gros, “Flipping-based Policy for Chance-Constrained Markov Decision Processes” Neural Information Processing Systems (NeurIPS), 2024.
10. Akifumi Wachi, Xun Shen, Yanan Sui, “A Survey of Constraint Formulations in Safe Reinforcement Learning” International Joint Conference on Artificial Intelligence (IJCAI), Survey Track, 2024.
11. Akifumi Wachi, Wataru Hashimoto, Kazumune Hashimoto, “Long-term Safe Reinforcement Learning with Binary Feedback” AAAI Conference on Artificial Intelligence (AAAI), 2024.
12. Xun Shen, Akifumi Wachi, Wataru Hashimoto, Kazumune Hashimoto, Shigemasa Takai “Safe Reinforcement Learning Using Model Predictive Control with Probabilistic Control Barrier Function” American Control Conference (ACC), 2024.
13. Akifumi Wachi, Wataru Hashimoto, Xun Shen, Kazumune Hashimoto, “Safe Exploration in Reinforcement Learning: A Generalized Formulation and Algorithms” Neural Information Processing Systems (NeurIPS), 2023.
14. Akifumi Wachi, Yunyue Wei, Yanan Sui, “Safe Policy Optimization with Local Generalized Linear Function Approximations” Neural Information Processing Systems (NeurIPS), 2021.
15. Ran Iwamoto, Ryosuke Kohita, Akifumi Wachi, “Polar Embedding” The SIGNLL Conference on Computational Natural Language Learning (CoNLL), 2021.
16. Daiki Kimura, Masaki Ono, Subhajit Chaudhury, Ryosuke Kohita, Akifumi Wachi, Don Joven Agravante, Michiaki Tatsubori, Asim Munawar, Alexander Gray, “Neuro-Symbolic Reinforcement Learning with First-Order Logic” Empirical Methods in Natural Language Processing (EMNLP), Short paper, 2021.
17. Ryosuke Kohita, Akifumi Wachi, Daiki Kimura, Subhajit Chaudhury, Michiaki Tatsubori, Asim Munawar, “Language-based General Action Template for Reinforcement Learning Agents” Association for Computational Linguistics (ACL), Findings, 2021.
18. Ryosuke Kohita, Akifumi Wachi, Yang Zhao, Ryuki Tachibana, “Q-learning with Language Model for Edit-based Unsupervised Summarization” Empirical Methods in Natural Language Processing (EMNLP), 2020
19. A. Wachi, Y. Sui, “Safe Reinforcement Learning in Constrained Markov Decision Processes”, International Conference on Machine Learning (ICML), 2020
20. A. Wachi, “Failure-Scenario Maker for Rule-based Agent using Multi-agent Adversarial Reinforcement Learning and its Application to Autonomous Driving”, International Joint Conference on Artificial Intelligence (IJCAI), 2019
21. A. Wachi, Y. Sui, Y. Yue, M. Ono, “Safe Exploration and Optimization of Constrained MDPs using Gaussian Processes”, AAAI Conference on Artificial Intelligence (AAAI), 2018
22. R. Sakagami, R. Takahashi, A. Wachi, Y. Koshiro, H. Maezawa, Y. Kasai, S. Nakasuka, “Integral Design Method for Simple and Small Mars Lander System Using Membrane Aeroshell”, Acta Astronautica.

23. A. Wachi, R. Takahashi, R. Sakagami, Y. Koshiro, Y. Kasai, S. Nakasuka, "Feasibility Study for Mars Landing with Small Space Probe", AIAA SPACE and Astronautics Forum and Exposition (2017 AIAA SPACE Forum), 2017.
24. A. Wachi, "Low-Thrust Trajectory Design to Improve Overall Mission Success Probability Incorporating Target Changes in Case of Engine Failures", 31st International Symposium on Space Technology and Science (ISTS), 2017. **Nominated to Best Student Paper Award: First Prize (General Chairperson Award)**
25. A. Wachi, N. Ozaki, S. Nakasuka, "Fault-Tolerant Low-Thrust Trajectory Design with Backups for Multiple Targets", 20th IFAC Symposium on Automatic Control in Aerospace (ACA), 2016. **Nominated to the Best Student Paper Award: First Prize.**

PATENTS

I have over 10 filed patents which can be found here:

- <https://patents.justia.com/search?q=akifumi+wachi>

ACADEMIC SERVICES

Conference Reviewing (Program Committee Member)

- ML: NeurIPS (2021 - 2025), ICML (2021 - 2026), ICLR (2022 - 2026), AISTATS (2024 - 2025)
- AI: IJCAI (2020 - 2025), AAAI (2020 - 2025)
- NLP: COLM (2024 - 2026), ACL (2020 - 2023), EMNLP (2020)

Journal Reviewing

- ACM Transactions on Evolutionary Learning and Optimization (Associate Editor)
- IEEE Transactions on Pattern Analysis and Machine Intelligence

AWARDS

Top Reviewer, NeurIPS 2024,

Dean's Award (second prize), from Faculty of Engineering, The University of Tokyo, 2018

Best Student Paper Award, First Prize (General Chairperson Award), Joint Conference: 31st ISTS, 26th ISSFD, and 8th NSAT, 2017, http://www.ists.or.jp/wp-content/uploads/2016/03/31_Prize-List.pdf

Best Student Paper Award (First Prize), IFAC Symposium on Automatic Control in Aerospace (ACA), 2016, <http://aca2016.ngcaerospace.com/studentaward.php>

FELLOWSHIP

Financial support to participate in Thirty-second AAAI conference on Artificial Intelligence by Society for Promotion of Space Science (SPSS), \$1.5K, February, 2018

Financial support to participate in International Astronautical Congress (IAC) by the Japan Society for Aeronautical and Space Sciences (JSASS), \$2.5K, September, 2017

Go Global Scholarships 2016 Short Study Abroad Scholarships: the University of Tokyo, \$1K/month, 3 months, 2016

Fellowship of the Leading Graduates Schools Program: Global Leader Program for Social Design and Management by the Ministry of Education, Culture, Sports, Science and Technology in Japan, \$1K/month,

1 year, 2016

Financial support to visit California Institute of Technology (Caltech) and NASA/JPL by the University of Tokyo, \$1.0K, February, 2014